

New and Practical Formulation for Overlaps of Bogoliubov Vacua

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New and Practical Formulation for Overlaps of Bogoliubov Vacua

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In this Letter, we present a new expression for the overlaps of wave functions in Hartree-Fock-Bogoliubov based theories. Starting from the Pfaffian formula by Bertsch *et al.* [1], an exact and computationally stable formula for overlaps is derived. We illustrate the convenience of this new formulation with a numerical application in the context of the particle-number projection method. This new formula allows for substantially increased precision and versatility in chemical, atomic, and nuclear physics applications, particularly for methods dealing with superfluidity, symmetry restoration, and uses of nonorthogonal many-body basis states.

研究目的

方法与验证

结论与意义

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HFB description

- Mean-field theories provide approximate solutions to the many-body Schrödinger equation
- **HFB method** describes the nuclear wave function as a product of independent quasiparticles.

$$\begin{pmatrix} h - \lambda & \Delta \\ -\Delta^* & -h^* + \lambda \end{pmatrix} \begin{pmatrix} U \\ V \end{pmatrix} = E \begin{pmatrix} U \\ V \end{pmatrix} \Rightarrow \begin{pmatrix} \beta_k^\dagger \\ \beta_k \end{pmatrix} = \begin{pmatrix} U^T & V^T \\ V^\dagger & U^\dagger \end{pmatrix} \begin{pmatrix} c_l^\dagger \\ c_l \end{pmatrix} \Rightarrow |\Phi\rangle = \prod_{k=1}^M \beta_k |0\rangle$$

Beyond mean-field

- Symmetry restoration and GCM introduce correlations beyond mean field.
- Require overlaps between **non-orthogonal HFB states**.

$$O_{KK'}^J = \langle \Phi | \hat{O} \hat{P}_{KK'}^J \hat{P}^N \hat{P}^Z | \Phi' \rangle = \frac{2J+1}{16\pi^3} \int_0^\pi e^{-i\phi_n N} d\phi_n \int_0^\pi e^{-i\phi_p Z} d\phi_p \int D_{MK}^{J*}(\Omega) d\Omega \langle \Phi | \hat{O} \hat{\mathcal{R}} | \Phi' \rangle$$

$$\langle \Phi | \hat{O} \hat{\mathcal{R}} | \Phi' \rangle = \langle \Phi | \hat{\mathcal{R}} | \Phi' \rangle \frac{\langle \Phi | \hat{O} \hat{\mathcal{R}} | \Phi' \rangle}{\langle \Phi | \hat{\mathcal{R}} | \Phi' \rangle}$$

Overlap: $\langle \Phi | \hat{\mathcal{R}} | \Phi' \rangle = \langle \Phi | \Phi'' \rangle$



- **Onishi formula :**

N. Onishi and S. Yoshida, [Nucl. Phys. 80, 367 \(1966\)](#).

$$\langle \Phi | \Phi' \rangle = \pm \sqrt{\det(U^\dagger U' + V^\dagger V')}$$

Sign problem

- **Robledo formula**

L. M. Robledo, [Phys. Rev. C 79, 021302\(R\) \(2009\)](#).

$$\langle \Phi | \Phi' \rangle = (-1)^{N(N+1)/2} \text{pf} \begin{pmatrix} M' & -1 \\ 1 & -M^* \end{pmatrix}, \quad M = (VU^{-1})^*$$

Invalid when U non-invertible

- **Bertsch&Robledo formula:**

G. F. Bertsch and L. M. Robledo, [Phys. Rev. Lett. 108, 042505 \(2012\)](#).

$$\langle \Phi | \Phi' \rangle = (-1)^{N/2} \frac{(\det C)^* \det C'}{\prod_{i,i'} v_i v'_{i'}} \text{pf} \begin{pmatrix} V^T U & V^T V'^* \\ -V'^\dagger V & U'^\dagger V'^* \end{pmatrix}$$

Small v makes results unstable

- **Other formula:**

K. Neergård and E. Wüst, [Nucl. Phys. A402, 311 \(1983\)](#).

Q. Haider and D. Gogny, [J. Phys. G 18, 993 \(1992\)](#).

F. Döna, [Phys. Rev. C 58, 872 \(1998\)](#).

M. Bender and P.-H. Heenen, [Phys. Rev. C 78, 024309 \(2008\)](#).

K. Hara, A. Hayashi, and P. Ring, [Nucl. Phys. A385, 14 \(1982\)](#).

M. Oi and N. Tajima, [Phys. Lett. B 606, 43 \(2005\)](#).

T. R. Rodríguez and J. L. Egido, [Phys. Rev. C 81, 064323 \(2010\)](#).

B. Bally and T. Duguet, [Phys. Rev. C 97, 024304 \(2018\)](#).

T. Mizusaki, M. Oi, and N. Shimizu, [Phys. Lett. B 779, 237 \(2018\)](#).

B. Avez and M. Bender, [Phys. Rev. C 85, 034325 \(2012\)](#).

K. Schmid, [Prog. Part. Nucl. Phys. 52, 565 \(2004\)](#).

Discard tiny v_i, v'_i levels

- Remove levels with $v_i, v'_i \sim 0$.

- **Disadvantage:** Need to verify that omitted levels have negligible contribution.

Introduce a cutoff parameter ϵ

- Replace small values by $v_i \rightarrow \epsilon$.
- **Disadvantage:** Results may depend on the choice of ϵ .

... *Z.-C. Gao, Q.-L. Hu, and Y. Chen, [Phys. Lett. B 732, 360 \(2014\)](#)*

$$\langle \Phi | \Phi' \rangle = (-1)^{N/2} \frac{(\det C)^* \det C'}{\prod_{i,i'} v_i v'_{i'}} \text{pf} \begin{pmatrix} V^T U & V^T V'^* \\ -V'^\dagger V & U'^\dagger V'^* \end{pmatrix}$$

Bloch-Messiah decomposition:

$$U = D \bar{U} C, V = D^* \bar{V} C$$

where D and C are both unitary matrices. $\bar{U}$ and $\bar{V}$ can be chosen as diagonal and skew-symmetric:

$$\bar{U} = \begin{pmatrix} 0 & & & & & & & \\ & \ddots & & & & & & \\ & & 0 & & & & & \\ & & & \boxed{\begin{smallmatrix} u_1 & 0 \\ 0 & u_1 \end{smallmatrix}} & & & & \\ & & & & \ddots & & & \\ & & & & & \boxed{\begin{smallmatrix} u_n & 0 \\ 0 & u_n \end{smallmatrix}} & & \\ & & & & & & 1 & \\ & & & & & & & \ddots \\ & & & & & & & & 1 \end{pmatrix}, \quad \bar{V} = \begin{pmatrix} 1 & & & & & & & \\ & \ddots & & & & & & \\ & & 1 & & & & & \\ & & & \boxed{\begin{smallmatrix} 0 & v_1 \\ -v_1 & 0 \end{smallmatrix}} & & & & \\ & & & & \ddots & & & \\ & & & & & \boxed{\begin{smallmatrix} 0 & v_n \\ -v_n & 0 \end{smallmatrix}} & & \\ & & & & & & 0 & \\ & & & & & & & \ddots \\ & & & & & & & & 0 \end{pmatrix}$$

Ring, The Nuclear many body problem (P247)

$$\langle \Phi | \Phi' \rangle = (-1)^{N/2} \frac{(\det C)^* \det C'}{\prod_{i,i'} v_i v'_{i'}} \text{pf} \begin{pmatrix} V^T U & V^T V'^* \\ -V'^\dagger V & U'^\dagger V'^* \end{pmatrix} = (-1)^{N/2} \frac{(\det C)^* \det C'}{\prod_{i,i'} v_i v'_{i'}} \text{pf}(M)$$

Apply Bloch-Messiah decomposition:

$$U = D \bar{U} C, V = D^* \bar{V} C$$

$$M = \begin{pmatrix} -C^T \bar{V} \bar{U} C & -C^T \bar{V} D^\dagger D' \bar{V}' C'^* \\ C'^\dagger \bar{V}' D'^T D^* \bar{V} C & C'^\dagger \bar{U}' \bar{V}' C'^* \end{pmatrix} = \begin{pmatrix} C^T & 0 \\ 0 & C'^\dagger \end{pmatrix} \begin{pmatrix} -\bar{V} \bar{U} & -\bar{V} D^\dagger D' \bar{V}' \\ \bar{V}' D'^T D^* \bar{V} & \bar{U}' \bar{V}' \end{pmatrix} \begin{pmatrix} C & 0 \\ 0 & C'^* \end{pmatrix}$$

Apply the relation:

$$\text{pf}(ABA^T) = \det(A) \text{pf}(B)$$

$$\text{pf}(M) = \det C \det C'^\dagger \text{pf} \begin{pmatrix} -\bar{V} \bar{U} & -\bar{V} D^\dagger D' \bar{V}' \\ \bar{V}' D'^T D^* \bar{V} & \bar{U}' \bar{V}' \end{pmatrix}$$

In order to avoid the numerical instability that arises in the computation of $\prod_{i,i'} v_i v'_{i'}$ we factorize it out of $\text{pf}(M)$.

Main point !

$$\begin{aligned} \text{pf}(M) &= \det C \det C'^{\dagger} \text{pf} \left[\begin{pmatrix} \bar{V}\Lambda^{-1} & 0 \\ 0 & \bar{V}'\Lambda'^{-1} \end{pmatrix} \begin{pmatrix} \Lambda\bar{U}\bar{V}^{-1}\Lambda & \Lambda D^{\dagger}D'\Lambda' \\ -\Lambda'D'^T D^*\Lambda & -\Lambda'\bar{V}'^{-1}\bar{U}'\Lambda' \end{pmatrix} \begin{pmatrix} -\Lambda^{-1}\bar{V} & 0 \\ 0 & -\Lambda'^{-1}\bar{V}' \end{pmatrix} \right] \\ &= \det C \det C'^{\dagger} \det(\bar{V}\Lambda^{-1}) \det(\bar{V}'\Lambda'^{-1}) \text{pf} \begin{pmatrix} \Lambda\bar{U}\bar{V}^{-1}\Lambda & \Lambda D^{\dagger}D'\Lambda' \\ -\Lambda'D'^T D^*\Lambda & -\Lambda'\bar{V}'^{-1}\bar{U}'\Lambda' \end{pmatrix}, \end{aligned}$$

where Λ is a diagonal matrix

$$\Lambda = \begin{pmatrix} \sqrt{v_0} & & & & \\ & \sqrt{v_0} & & & \\ & & \sqrt{v_1} & & \\ & & & \ddots & \\ & & & & \sqrt{v_{N/2-1}} \end{pmatrix} \quad \bar{V}\Lambda^{-1} = \begin{pmatrix} 0 & \sqrt{v_0} & & & \\ -\sqrt{v_0} & 0 & & & \\ & & \ddots & & \\ & & & 0 & \sqrt{v_n} \\ & & & -\sqrt{v_n} & 0 \\ & & & & \ddots & \\ 0 & & & & & & 0 \end{pmatrix}$$

Because $\det(\bar{V}\Lambda^{-1}) = \prod_i v_i$

$$\langle \Phi | \Phi' \rangle = (-1)^{N/2} \frac{(\det C)^* \det C'}{\prod_{i,i'} v_i v'_{i'}} \text{pf}(M) = (-1)^{N/2} \text{pf} \begin{pmatrix} \Lambda\bar{U}\bar{V}^{-1}\Lambda & \Lambda D^{\dagger}D'\Lambda' \\ -\Lambda'D'^T D^*\Lambda & -\Lambda'\bar{V}'^{-1}\bar{U}'\Lambda' \end{pmatrix}$$

$$\begin{aligned} \text{pf}(M) &= \det C \det C'^{\dagger} \text{pf} \left[\begin{pmatrix} \bar{V}\Lambda^{-1} & 0 \\ 0 & \bar{V}'\Lambda'^{-1} \end{pmatrix} \begin{pmatrix} \Lambda\bar{U}\bar{V}^{-1}\Lambda & \Lambda D^{\dagger}D'\Lambda' \\ -\Lambda'D'^T D^*\Lambda & -\Lambda'\bar{V}'^{-1}\bar{U}'\Lambda' \end{pmatrix} \begin{pmatrix} -\Lambda^{-1}\bar{V} & 0 \\ 0 & -\Lambda'^{-1}\bar{V}' \end{pmatrix} \right] \\ &= \det C \det C'^{\dagger} \det(\bar{V}\Lambda^{-1}) \det(\bar{V}'\Lambda'^{-1}) \text{pf} \begin{pmatrix} \Lambda\bar{U}\bar{V}^{-1}\Lambda & \Lambda D^{\dagger}D'\Lambda' \\ -\Lambda'D'^T D^*\Lambda & -\Lambda'\bar{V}'^{-1}\bar{U}'\Lambda' \end{pmatrix}, \end{aligned}$$

where Λ is a diagonal matrix

$$\Lambda = \begin{pmatrix} \sqrt{v_0} & & & & \\ & \sqrt{v_0} & & & \\ & & \sqrt{v_1} & & \\ & & & \ddots & \\ & & & & \sqrt{v_{N/2-1}} \end{pmatrix} \quad \bar{V}\Lambda^{-1} = \begin{pmatrix} 0 & \sqrt{v_0} & & & \\ -\sqrt{v_0} & 0 & & & \\ & & \ddots & & \\ & & & 0 & \sqrt{v_n} \\ & & & -\sqrt{v_n} & 0 \\ & & & & \ddots & \\ 0 & & & & & & 0 \end{pmatrix}$$

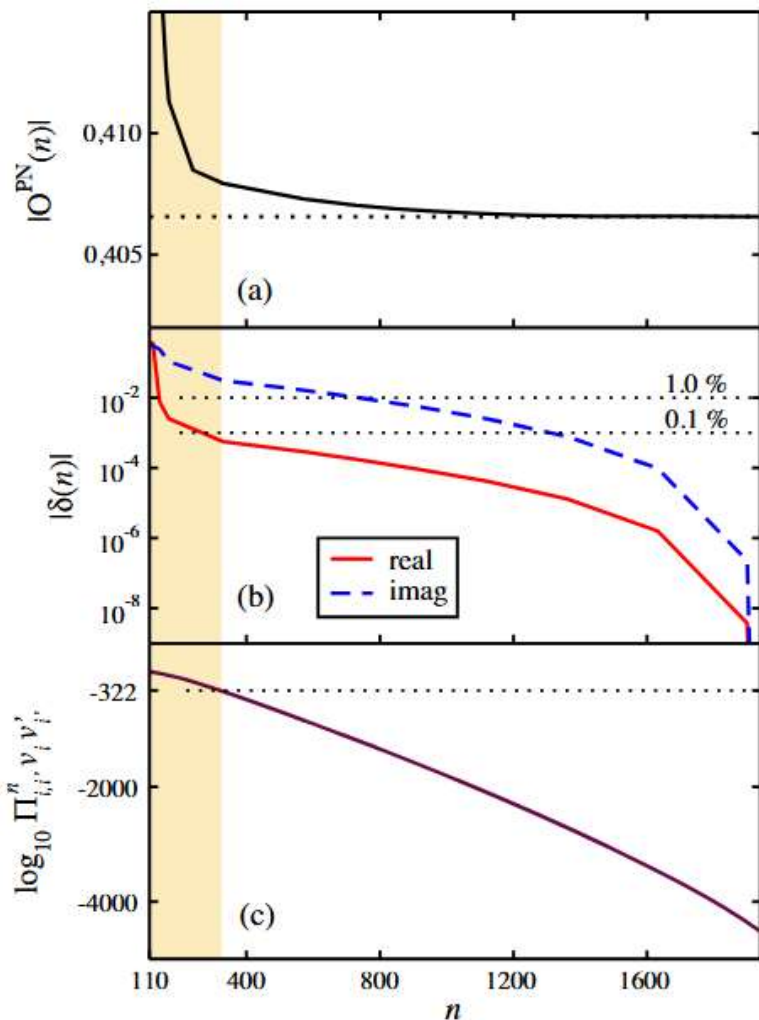
Because $\det(\bar{V}\Lambda^{-1}) = \prod_i v_i$

$$\langle \Phi | \Phi' \rangle = (-1)^{N/2} \frac{(\det C)^* \det C'}{\prod_{i,i'} v_i v_{i'}} \text{pf}(M) = (-1)^{N/2} \text{pf} \begin{pmatrix} -\bar{U}\sigma & \Lambda D^{\dagger}D'\Lambda' \\ -\Lambda'D'^T D^*\Lambda & \sigma\bar{U}' \end{pmatrix}$$

Example



$$O^{PN} = \langle \Phi | e^{-i\hat{N}\phi} | \Phi \rangle = \langle \Phi | \Phi' \rangle = (-1)^{n/2} \text{pf} \begin{pmatrix} [-\bar{U}\sigma]_{n \times n} & [\Lambda D^\dagger D' \Lambda']_{n \times n'} \\ [-\Lambda' D'^T D^* \Lambda]_{n' \times n} & [\sigma \bar{U}']_{n' \times n'} \end{pmatrix}$$



- HFB calculations of ^{192}Pb using SLY4 functional, with 17 major shell and separable Gaussian pairing interaction.
- $U' = Ue^{i\phi}, V' = Ve^{-i\phi}$ with $\phi = \pi/10$

- ✓ **Convergence:** The value of the matrix elements smoothly converges to the exact value as n is increased.
- ✓ **Accuracy:** A relative error less than 1% can be achieved with around 800 states.
- ✓ **Numerical stability:** The computation becomes unstable when the prefactor $\prod_i^n v_i v'_i$ falls below 10^{-322} in double precision. In such case, $n = 323$ and $v_i = 0.097593$.

- The new formula allows all states to be included **without encountering any numerical instability**.
- The new formulation **adds no extra computational effort or complexity** and allows **systematic truncation with smooth convergence toward the exact value**.
- The formulation opens the possibility for **precision calculations of nuclear reactions and nuclear structure** by exploiting symmetry breaking.